

Data-Driven Fault-Tolerant Control for Attitude Synchronization of Nonlinear Quadrotors

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Abstract—In this article, the data-driven fault-tolerant synchronization control problem is investigated for unknown cooperative quadrotors subject to nonlinearities and multiple actuator faults in the quadrotor dynamics. A distributed observer is provided to estimate the state of the virtual leader. Based on the reinforcement learning theory, the optimal control policy is learned for each quadrotor without any knowledge of the quadrotor dynamic information. Then, the learned control policy is used to construct a data-based fault-tolerant controller to restrain the effects of quadrotor actuator faults. Stability of the constructed controller is proven and the simulation results illustrate the effectiveness of the proposed controller.

Index Terms—Data-based, fault-tolerant control (FTC), quadrotor system, reinforcement learning (RL), synchronization control.

I. INTRODUCTION

In recent years, the unmanned aerial vehicles (UAVs) have witnessed much progress in military and civilian areas. As a typical class of UAVs, quadrotors have received much research interest because of their capabilities of hovering, slow flight, and vertical take-off and landing (see [1]). Furthermore, the cooperation of the multiagent quadrotor system exhibits special superiority in the complicated task such as telecommunication relay, cooperative load transportation, and wilderness search (see [2]). The quadrotors are composed of a rigid frame with four individual rotors, which generate the control torques for the quadrotor by changing the spinning speeds of the four rotors. However, the actuating rotors are prone to faults because of the degradation of the motors or the damage of the propellers (see, e.g., [3]). The appearance of the actuator faults may result in the undesirable trajectory tracking performance, instability of the multiagent quadrotor system, and even catastrophic consequence (see [4]). Therefore, it is necessary to address the fault-tolerant control (FTC) problem for the multiagent quadrotor system under actuator faults.

Recently, some FTC methods have been implemented to the cooperative control of the multiagent systems. In [5], the distributed adaptive

FTC problem was studied for the synchronization of linear multiagent systems under the effects of actuator faults. In [6], the distributed FTC problem was investigated to achieve the formation for second-order linear multiagent systems in the presence of multiple actuator faults. In [7], a fault-tolerant cooperative controller for a team of nonlinear systems was developed by using the integral sliding mode control (SMC) technique. The fault-tolerant cooperative control problem for the multi-UAV system has been studied in [8]–[11]. In [8], a fault-tolerant formation controller was studied for multiple UAVs by considering the health conditions of the UAV team and compensate the effects of the actuators faults through the healthy/redundant actuators. In the robust controller design of [9], multiple actuator faults were considered in the equivalent disturbances for multiple cooperative tail-sitters and a robust compensator was designed to restrain the actuator faults. In [10], a distributed fault-tolerant controller was studied to achieve the desired synchronization for multiple cooperative UAVs with nonlinearities and actuator faults in the dynamics. In [11], a fault-tolerant formation controller involving a decentralized state observer and an adaptive fault estimator was developed for nonlinear quadrotor team to drive each quadrotor to the desired formation pattern. However, the fault-tolerant synchronization controller design of multiple quadrotors is still challenging in three aspects.

First, the quadrotor is a multiple-input multiple-output system with high nonlinearities and strong couplings in the attitude dynamics, as illustrated in [12] and [13]. Some traditional optimal control methods were applied to the FTC of the robotic system under actuator faults, as shown in [14] and [15]. But, the optimal control policies in [14] and [15] were obtained by directly solving linear algebraic Riccati equations, which is unpractical for the highly nonlinear quadrotor attitude system. To obtain an optimal control policy for the nonlinear quadrotor system, a highly nonlinear Hamilton–Jacobi–Bellman (HJB) equation is required to be solved for the traditional optimal control methods, as shown in [16]. However, it is difficult to obtain the optimal control policy by directly solving the nonlinear HJB equation for quadrotor systems. Second, as depicted in [17], the quadrotor is suffering from multiple simultaneous actuator faults in practical applications. The probability of the multiquadrotor system encountering multiple simultaneous actuator faults is growing due to the increasing team scale and the complexity of the multiquadrotor system (see [18]). Besides, the actuator faults are different, resulting in different dynamics, for different quadrotors in the team. Therefore, the global multiagent dynamical system is actually heterogenous. It is difficult to address the multiple simultaneous actuator faults for multiple quadrotors with different dynamics. Third, the accurate dynamic model of the quadrotor system with the nonlinearities and couplings is difficult to obtain in practical applications. It should be noted that most existing traditional FTC methods such as the sliding mode control method in [19], the robust control method in [9], and the adaptive control methods in [5], [6], and [11] are model-dependent and focus on matching the compensated systems to the predefined reference models, as illustrated in [20]. The inaccurate quadrotor dynamic model will result in the deterioration of the flight performance for the

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model-dependent controller, which brings challenge for the traditional control methods.

Due to the aforementioned challenges, it is important to design a data-driven fault-tolerant synchronization controller for a highly nonlinear multi-quadrotor system under multiple simultaneous actuator faults without requirement of the knowledge of the quadrotor dynamics. In the latest years, reinforcement learning (RL) has become a powerful data-driven approach to learn from experience the optimal control policy and emerged in the model-free optimal controller design for quadrotors. In [21], the RL technique was used to learn a swing-free optimal policy for a quadrotor with suspended payloads. In [22], an RL-based path tracking controller and an RL-based attitude controller were studied for a quadrotor to achieve the complex construction tasks. In [23], the optimal position controller and the optimal attitude controller were learned by using the RL approaches. However, the works in [21]–[23] mainly focused on the optimal control of a single-quadrotor system without actuator faults and the optimal control for quadrotors with multiple actuator faults was not studied further. In [20], a series of RL algorithms were implemented in the data-driven optimal FTC design for a single quadrotor to estimate and compensate the actuator faults. However, the data-driven FTC problem for the synchronization of multiple nonlinear and coupled quadrotor systems subject to multiple actuator faults has not been discussed in the literatures mentioned before.

Therefore, the fault-tolerant synchronization control problem for multiple unknown quadrotor systems with nonlinearities, couplings, and multiple actuator faults in the dynamics remains an open issue. In this article, a distributed active fault-tolerant synchronization controller is designed including a distributed observer to estimate the state of the virtual leader and an adaptive fault-tolerant controller to achieve the desired synchronization. The distributed observer is constructed to estimate the leader state using the state information of its neighbors and itself. Combining the optimal control theory and the RL approach, an optimal control policy is learned for the quadrotor attitude nominal system without knowledge of the dynamic information. Then, the adaptive fault-tolerant controller involving the learned control policy is designed to compensate the actuator faults. Therefore, the proposed fault-tolerant synchronization controller can achieve the model-free optimal control of the nonlinear quadrotor nominal system and restrain the effects of the multiple simultaneous actuator faults in the quadrotor system, simultaneously. Compared to previous studies, the features of the proposed data-driven fault-tolerant synchronization controller for cooperative quadrotors are three-fold.

First, the RL-based FTC approach for a single-quadrotor system is extended to the fault-tolerant synchronization control for the cooperative quadrotor team. The proposed fault-tolerant controller is not only optimal for the quadrotor nominal system with nonlinear and couple dynamics but also tolerant to multiple simultaneous actuator faults. But the works in [20]–[23] only discussed the RL-based optimal control for a single-quadrotor system and the fault-tolerant cooperative control of multi-quadrotor systems was not further discussed. Second, the quadrotors are considered as multi-input multi-output (MIMO) continuous-time systems with high nonlinearities and strong couplings in the dynamics. But the FTC problem for nonlinear and coupled systems was not discussed in [5] and [6]. Third, by using RL approach, the control parameters of the fault-tolerant controller are updated based on the collected data without any knowledge of the quadrotor dynamics. But the dynamic information of the systems was required in the fault-tolerant controller design of [5]–[8], [10], and [11].

The rest of this article are given as follows. Section II gives the preliminaries on the graph theory, the quadrotor model, the actuator fault model, and the problem formulation. The fault-tolerant synchronization

controller is designed in Section III and the stability analysis is given. In Section IV, the simulation results of six cooperative quadrotor are shown and the conclusion is given in Section V.

II. PRELIMINARIES

A. Graph Theory

Define $G = (V(G), E(G), W(G))$ as a time invariant graph, where $V(G) = \{v_1, v_2, \dots, v_N\}$ indicates the vertex set, $E(G) \subset V(G) \times V(G)$ indicates the edge set, and $W(G) = [w_{ij}] \in \mathbb{R}^{N \times N}$ indicates the associated adjacency matrix. Let $\Phi = \{1, 2, \dots, N\}$. For v_i and v_j ($i, j \in \Phi$), $w_{ij} > 0$ if and only if $(v_i, v_j) \in E$ and otherwise, $w_{ij} = 0$. The neighbors of v_i is denoted as $N_i = \{v_j | (v_i, v_j) \in E\}$. Define d_i as the in-degree of v_i with $d_i = \sum_{j=1}^N w_{ij}$. Let $L = D(G) - W(G)$, where $D(G) = \text{diag}(d_i) \in \mathbb{R}^{N \times N}$. A path form v_i to v_j is a sequence of ordered edges in the form $\{(v_i, v_m), (v_m, v_n), \dots, (v_k, v_j)\}$. Define M as a diagonal matrix with $M = \text{diag}(\gamma_i) \in \mathbb{R}^{N \times N}$, where γ_i is a constant satisfies that $\gamma_i > 0$ if quadrotor i is connected to the leader, otherwise, $\gamma_i = 0$. The graph G is said to have a spanning tree if there exist at least one vertex connected to all the other vertices through a path.

B. Nominal Quadrotor Model

Each of the N quadrotors is modeled as a rigid body with four rotors. Denote $\Theta_i = [\phi_i \ \theta_i \ \psi_i]^T \in \mathbb{R}^{3 \times 1}$ as the Euler angle of quadrotor i . The attitude dynamic model of quadrotor i is described as (see [13])

$$J_{\Theta_i} \ddot{\Theta}_i = -C(\Theta_i, \dot{\Theta}_i) \dot{\Theta}_i + \tau_i \quad (1)$$

where $J_{\Theta_i} = \text{diag}\{J_{\phi_i}, J_{\theta_i}, J_{\psi_i}\} \in \mathbb{R}^{3 \times 3}$ indicates the inertial matrix, $\tau_i = [\tau_{\phi_i} \ \tau_{\theta_i} \ \tau_{\psi_i}]^T \in \mathbb{R}^{3 \times 1}$ indicates the control torque, and $C(\Theta_i, \dot{\Theta}_i) \in \mathbb{R}^{3 \times 3}$ indicates the highly nonlinear and coupled Coriolis term shown in [24]. Define $\omega_{i,Rj}$ ($i \in \Phi, j = 1, 2, 3, 4$) as the spinning speed of the j th rotor of quadrotor i . The control torque τ_i is given by $\tau_{\phi_i} = k_{li} k_{\omega_i} u_{\phi_i}$, $\tau_{\theta_i} = k_{li} k_{\omega_i} u_{\theta_i}$, and $\tau_{\psi_i} = k_{\tau_i} u_{\psi_i}$, $i \in \Phi$, where k_{li} is the distance between the mass center of quadrotor i and the rotor and k_{ω_i}, k_{τ_i} are the positive scaling factors. $u_{\phi_i}, u_{\theta_i}, u_{\psi_i}$ are the control inputs satisfying that $u_{\phi_i} = \omega_{i,R2}^2 - \omega_{i,R4}^2$, $u_{\theta_i} = \omega_{i,R1}^2 - \omega_{i,R3}^2$, and $u_{\psi_i} = \sum_{j=1}^4 (-1)^{j+1} \omega_{i,Rj}^2$. Let $u_{zi} = \sum_{j=1}^4 \omega_{i,Rj}^2$, which generates the thrust for quadrotor i to restrain the effects of the gravity. Due to the existence of a power distribution board, the control inputs $u_{\phi_i}, u_{\theta_i}, u_{\psi_i}, u_{zi}$ can be distributed to the four rotors to control the rotational motions of quadrotor i . Define $b_{\tau_i} = \text{diag}(b_{\tau_i,1}, b_{\tau_i,2}, b_{\tau_i,3}) = J_{\Theta_i}^{-1} \text{diag}(k_{li} k_{\omega_i}, k_{li} k_{\omega_i}, k_{\tau_i})$ ($i \in \Phi$). Then, the quadrotor model (1) can be rewritten as

$$\ddot{\Theta}_i = -J_{\Theta_i}^{-1} \left(C(\Theta_i, \dot{\Theta}_i) \dot{\Theta}_i \right) + b_{\tau_i} u_{\tau_i} \quad (2)$$

where $u_{\tau_i} = [u_{\phi_i} \ u_{\theta_i} \ u_{\psi_i}]^T \in \mathbb{R}^{3 \times 1}$.

Remark 1: In practical applications, the inertial matrix J_{Θ_i} is difficult to be known due to different payloads, the quadrotor carries in each flight test. Furthermore, the input matrix b_{τ_i} may vary because of the changes of air density or the type of the propellers.

In the following sections, a data-driven fault-tolerant synchronization controller will be provided to achieve the attitude synchronization of multiple quadrotors without requirement of the quadrotor dynamic information.

C. Actuator Fault Model

In practical flight tasks of multiple cooperative quadrotors, the structure damage to the propellers or the accidental change of the physical

parameters in the quadrotor rotors may lead to a constant partial loss of effectiveness (LOE) in the actuator. Similarly to [17], the actuator fault for the j th rotor of quadrotor i is given by

$$\omega_{i,Rj}^{2*} = \chi_{ij} \omega_{i,Rj}^2, i \in \Phi, j = 1, 2, 3, 4 \quad (3)$$

where $\omega_{i,Rj}^{2*}$ is the square of the actual spinning speed, $\omega_{i,Rj}^2$ is the square of the commanded spinning speed, and χ_{ij} is an unknown positive constant with $0 < \chi_{ij} \leq 1$. The case of $\chi_{ij} = 1$ indicates that the j th rotor of quadrotor i is healthy. It should be noted that $\chi_{ij} = 0$ indicates that the actuator is stuck and redundant actuators are required to recover the flight performance, which is difficult for quadrotors with no actuator redundancy. Therefore, this article only focuses on the partial LOE faults for quadrotors due to the lack of actuator redundancy. For quadrotor i , define K_{χ_i} as the actuator fault matrix with $K_{\chi_i} = \text{diag}\{\chi_{i1}, \chi_{i2}, \chi_{i3}, \chi_{i4}\}$. In practical applications, the LOE actuator faults of each quadrotor in the quadrotor team may not occur at the same time. Then, one can obtain the dynamic model of quadrotor i with the actuator faults as

$$\ddot{\Theta}_i = -J_{\Theta_i}^{-1} \left(C(\Theta_i, \dot{\Theta}_i) \dot{\Theta}_i \right) + b_{\tau_i} (u_{\tau_i} - \rho(t - t_{si}) u_{\Delta i}) \quad (4)$$

where $\rho(t - t_{si})$ is a step function and t_{si} is a fault starting time of quadrotor i satisfying that $\rho(t - t_{si}) = 1$ if $t > t_{si}$ and otherwise, $\rho(t - t_{si}) = 0$. $u_{\Delta i}$ in (4) is the input uncertainty resulted from the actuator faults of the rotors with

$$u_{\Delta i} = b_{\tau_i}^{-1} J_{\Theta_i}^{-1} G_m (I_4 - K_{\chi_i}) \bar{\omega}_i^2 \quad (5)$$

where $\bar{\omega}_i^2 = [\omega_{i,R1}^2, \omega_{i,R2}^2, \omega_{i,R3}^2, \omega_{i,R4}^2]^T$, $I_N \in \mathbb{R}^{N \times N}$ is a unit matrix, and G_m is the transformation matrix with

$$G_m = \begin{bmatrix} 0 & k_{li} k_{\omega i} & 0 & -k_{li} k_{\omega i} \\ k_{li} k_{\omega i} & 0 & -k_{li} k_{\omega i} & 0 \\ k_{\tau i} & -k_{\tau i} & k_{\tau i} & -k_{\tau i} \end{bmatrix}. \quad (6)$$

Remark 2: It can be seen from (4) to (6) that the LOE actuator faults of the rotors lead to the control input uncertainty of the quadrotor control system. In practical flight of the cooperative quadrotor team, the spinning speed $\bar{\omega}_i^2$ ($i \in \Phi$) is bounded but changes all the time. Therefore, the input uncertainty $u_{\Delta i}$ is bounded and time-varying with $\|u_{\Delta i}\| \leq \bar{u}_{\Delta i}$. The bound of the input uncertainty $\bar{u}_{\Delta i}$ cannot be directly obtained in practical applications and is estimated in this article.

D. Problem Formulation

The objective of this article is to design a data-driven fault-tolerant controller for a cooperative nonlinear quadrotor team to achieve the attitude synchronization under the LOE actuator faults. The proposed fault-tolerant controller is required to learn the optimal control policy for the quadrotor nominal system and restrain the LOE actuator faults without knowledge of the quadrotor dynamic information. The quadrotors in this article is connected with each other by a communication graph G satisfying Assumption 1.

Assumption 1: There exists more than one spanning tree G in the communication graph of quadrotors.

Assumption 1 is a traditional assumption in the synchronization design of multiagent systems. Let $x_i = [\Theta_i^T, \dot{\Theta}_i^T]^T \in \mathbb{R}^{6 \times 1}$ be the state of quadrotor i . From (4), one can rewrite the quadrotor model as the following:

$$\dot{x}_i = f_{\Theta_i}(x_i) + B_{\tau_i} (u_{\tau_i} - \rho(t - t_{si}) u_{\Delta i}) \quad (7)$$

where $B_{\tau_i} = [0_{3 \times 3} \ b_{\tau_i}]^T \in \mathbb{R}^{6 \times 3}$ and $f_{\Theta_i}(x_i)$ is the nonlinear term with

$$f_{\Theta_i}(x_i) = \begin{bmatrix} 0_{3 \times 3} & I_3 \\ 0_{3 \times 3} & -J_{\Theta_i}^{-1} C(\Theta_i, \dot{\Theta}_i) \end{bmatrix} x_i. \quad (8)$$

Define $x_0 = [\Theta_0^T, \dot{\Theta}_0^T]^T \in \mathbb{R}^{6 \times 1}$ as the state of the virtual leader.

Remark 3: In the distributed synchronization of multiquadrotor systems, the state of the virtual leader x_0 is only directly available for the quadrotors connected to the leader, i.e., the neighbors of the leader, and thus, unavailable for all the following quadrotors. In the following section, a distributed observer is designed to estimate the state of the virtual leader for each quadrotor.

III. CONTROL PROTOCOL DESIGN

A. Distributed Observer

By using the states of the neighbors, a distributed observer for quadrotor i to estimate the leader state is designed as

$$\dot{\varsigma}_i = A_{\Theta_0} \varsigma_i + v \sum_{j \in N_i} w_{ij} (\varsigma_j - \varsigma_i) + \gamma_i (x_0 - \varsigma_i) \quad (9)$$

where $\varsigma_i \in \mathbb{R}^{6 \times 1}$ is the state of the observer for quadrotor i and $A_{\Theta_0} \in \mathbb{R}^{6 \times 6}$ is the dynamic matrix of the virtual leader. By using the distributed observer in (9) to estimate the leader states for each quadrotor, the complicated synchronization control task for the quadrotor team can be distributed into a trajectory tracking task for each quadrotor. Then, the attitude synchronization of quadrotors can be achieved by independently designing trajectory tracking controller for each quadrotor in the team. Define $\lambda_i(a)$ as the i th eigenvalue of the matrix a and $\lambda_R(a)$ a set of the real parts of the eigenvalues of a . The convergence of the estimated state ς_i in (9) to the leader state x_0 is shown in Theorem 1.

Theorem 1: Assume that the communication of the multiquadrotor system satisfies Assumption 1. Then, the state of the observer ς_i in (9) will exponentially converge to the state of the virtual leader x_0 , if the positive scale v satisfies that $v > \max(\lambda_R(A_{\Theta_0})) / \min(\lambda_R(L + M))$.

Proof: Let $\tilde{x}_{\Theta_i} = [\tilde{x}_{\phi i}, \dot{\tilde{x}}_{\phi i}, \tilde{x}_{\theta i}, \dot{\tilde{x}}_{\theta i}, \tilde{x}_{\psi i}, \dot{\tilde{x}}_{\psi i}]^T \in \mathbb{R}^{6 \times 1}$ ($i \in \Phi$) be the estimation error of the observer for quadrotor i with

$$\tilde{x}_{\Theta_i} = \varsigma_i - x_0. \quad (10)$$

By differentiating (10) and using (9), one can obtain that

$$\dot{\tilde{x}}_{\Theta_i} = A_{\Theta_0} (\varsigma_i - x_0) + v \sum_{j \in N_i} w_{ij} (\varsigma_j - \varsigma_i) + \gamma_i (x_0 - \varsigma_i). \quad (11)$$

Then, one can derive the global error dynamics as

$$\dot{\tilde{x}}_{\Theta} = P_0 \tilde{x}_{\Theta} \quad (12)$$

where $\tilde{x}_{\Theta} = [\tilde{x}_{\Theta 1}^T, \tilde{x}_{\Theta 2}^T, \dots, \tilde{x}_{\Theta N}^T]^T$ and $P_0 = I_N \otimes A_{\Theta_0} - v(L + M) \otimes I_6$. It can be observed from (12) that the estimation error $\tilde{x}_{\Theta}$ will converge to 0 if P_0 is Hurwitz. The eigenvalues of the matrix P_0 are given as

$$\{\lambda_i(A_{\Theta_0}) - v\lambda_j(L + M) | i = 1, 2, \dots, 6, j = 1, 2, \dots, N\}. \quad (13)$$

Since the communication graph G has a spanning tree, one can see that all the eigenvalues of the matrix $L + M$ have positive real parts. Then, all the real parts of the eigenvalues of the matrix P_0 are negative

if v satisfies that $v > \max(\lambda_R(A_{\Theta 0}))/\min(\lambda_R(L + M))$, i.e., P_0 is Hurwitz if v satisfies that $v > \max(\lambda_R(A_{\Theta 0}))/\min(\lambda_R(L + M))$. ■

B. Fault-Tolerant Controller Design

In the following, a fault-tolerant controller is proposed to restrain multiple simultaneous actuator faults in the quadrotor team. By combining the estimated leader state with the quadrotor state, one can construct a quadrotor augmented system from (7) and (9) as

$$\dot{X}_i = F_{\Theta i}(X_i) + \bar{B}_i(u_{\tau i} + \rho(t - t_{si})u_{\Delta i}) + M_i \varepsilon_i \quad (14)$$

where $X_i = [x_i^T \ \varsigma_i^T]^T$, $F_{\Theta i}(X_i) = [f_{\Theta i}(x_i)^T \ \varsigma_i^T A_{\Theta 0}^T]^T$, $\bar{B}_i = [B_{\tau i}^T \ 0]^T$, $M_i = [0 \ -v]^T$, and $\varepsilon_i = \sum_{j \in N_i} w_{ij}(\varsigma_i - \varsigma_j) + \gamma_i(\varsigma_i - x_0)$. ε_i will converge to 0, if the positive scale v is selected properly according to Theorem 1. The quadrotor system in (14) can be recognized as a nominal augmented system with the input uncertainty resulted from the actuator faults. By ignoring the input uncertainty, one can design the performance function for the nominal augmented system in (14) as

$$V_i(X_i) = \int_0^\infty e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + u_{\tau i}^T R_{\Theta i} u_{\tau i} d\tau \quad (15)$$

where $Q_{\Theta i} > 0$, $R_{\Theta i} > 0$, $e_{\Theta i} = [e_{\phi i} \ e_{\theta i} \ e_{\psi i}]^T \in \mathbb{R}^{3 \times 1}$ is the attitude tracking error with $e_{\Theta i} = [C_{\Theta} \ -C_{\Theta}]X_i$ and $C_{\Theta} = [I_3 \ 0_{3 \times 3}]$. Differentiating (15) and considering the nominal augmented system in (14) yield the following Bellman equation:

$$e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + (u_{\tau i})^T R_{\Theta i} u_{\tau i} + (\Delta V_i)^T [F_{\Theta i}(X_i) + M_i \varepsilon_i + \bar{B}_i u_{\tau i}] = 0. \quad (16)$$

According to the traditional optimal control theory (see [25]), one can obtain the optimal control law $u_{\tau i}^* = -\frac{1}{2}R_{\Theta i}^{-1}\bar{B}_i^T \Delta V_i^*$ to minimize the performance function (15) by solving the following HJB equation:

$$e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + (\Delta V_i^*)^T F_{\Theta i} + (\Delta V_i^*)^T M_i \varepsilon_i - \frac{1}{4}(\Delta V_i^*)^T \bar{B}_i R_{\Theta i}^{-1} \bar{B}_i^T \Delta V_i^* = 0 \quad (17)$$

where V_i^* indicates an optimal performance function to be solved from (17) and $\Delta V_i^* = \partial V_i^* / \partial X_i$. It should be noted that the control input $u_{\tau i}^*$ can only guarantee the optimality of the nominal augmented system without actuator faults. In order to achieve the synchronization control of the quadrotor team under the LOE actuator faults, the fault-tolerant controller for each quadrotor is constructed as

$$u_{\tau i}(t) = \frac{u_{\tau i}^*(x)}{2} + \frac{\hat{u}_{\Delta i}^2 R_{\Theta i} u_{\tau i}^*(x)}{\|R_{\Theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta i} + \sigma(t)} \quad (18)$$

where $\sigma(t)$ is a positive bounded function and satisfies that

$$\int_t^\infty \sigma(t) d\tau \leq \bar{\sigma} < \infty. \quad (19)$$

$\hat{u}_{\Delta i}$ is the estimated value of the bound $\bar{u}_{\Delta i}$ with

$$\dot{\hat{u}}_{\Delta i} = -k_r \sigma(t) \hat{u}_{\Delta i} + 2k_r \|R_{\Theta i} u_{\tau i}^*(x)\| \quad (20)$$

where k_r is a positive constant. It can be observed from (18) that the fault-tolerant controller depends on the optimal control law $u_{\tau i}^*$, which can be obtained by using the RL algorithm in the following section.

C. Data-Driven RL Algorithm for Quadrotor Team

It should be noted that the HJB equation (17) is nonlinear with respect to V_i^* , and thus, the solution is difficult to obtain. Furthermore, in practical synchronization of quadrotor team, the accurate and complete quadrotor dynamic information in (17) is unavailable. In the following, an RL algorithm is designed to learn the solution to (17) without any knowledge of the dynamic information of the quadrotor. Denote $V_i^n(X_i)$ and $u_{\tau i}^n$ as the updated performance function and control policy in the n th iteration. Consider the augmented system (14) explored by an original stabilizing control policy u_{0i} with a bounded noise $u_{ei} \in \mathbb{R}^{3 \times 1}$ as

$$\dot{X}_i = F_{\Theta i}(X_i) + \bar{B}_i u_{\tau i}^n + M_i \varepsilon_i + \bar{B}_i(u_{0i} + u_{ei} - u_{\tau i}^n), i \in \Phi. \quad (21)$$

Lemma 1: For any bounded exploring noise u_{ei} , the updated performance function $V_i^n(X_i)$ satisfies the following Bellman equation:

$$\begin{aligned} & V_i^n(X_i(t + \Delta t)) - V_i^n(X_i(t)) \\ &= \int_t^{t+\Delta t} -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} - (u_{\tau i}^n)^T R_{\Theta i} u_{\tau i}^n d\tau \\ & \quad - \int_t^{t+\Delta t} 2(u_{\tau i}^{n+1})^T R_{\Theta i} (u_{0i} + u_{ei} - u_{\tau i}^n) d\tau. \end{aligned} \quad (22)$$

Proof: By differentiating (15) and using (16), one can obtain that

$$\begin{aligned} \dot{V}_i^n &= (\Delta V_i^n)^T (F_{\Theta i}(X_i) + \bar{B}_i u_{\tau i}^n + M_i \varepsilon_i) \\ & \quad + (\Delta V_i^n)^T \bar{B}_i (u_{0i} + u_{ei} - u_{\tau i}^n) \\ &= -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} - (u_{\tau i}^n)^T R_{\Theta i} u_{\tau i}^n \\ & \quad - 2(u_{\tau i}^{n+1})^T R_{\Theta i} (u_{0i} + u_{ei} - u_{\tau i}^n). \end{aligned} \quad (23)$$

By integrating (23) from t to $t + \Delta t$, one has that

$$\begin{aligned} & V_i^n(X_i(t + \Delta t)) - V_i^n(X_i(t)) \\ &= \int_t^{t+\Delta t} -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} - (u_{\tau i}^n)^T R_{\Theta i} u_{\tau i}^n d\tau \\ & \quad - \int_t^{t+\Delta t} 2(u_{\tau i}^{n+1})^T R_{\Theta i} (u_{0i} + u_{ei} - u_{\tau i}^n) d\tau. \end{aligned} \quad (24)$$

Then, one can have (22). ■

Remark 4: It can be seen that the quadrotor dynamics is not required in (22) and the performance function V_i^n and the control policy $u_{\tau i}^{n+1}$ in the n th iteration can be updated by solving (22) for V_i^n and $u_{\tau i}^{n+1}$, which yields a data-driven RL algorithm in Algorithm 1.

The convergence of Algorithm 1 can be proven similarly to the lines of reasoning of [26] and is omitted here. The flowchart of Algorithm 1 is depicted in Fig. 1. It can be observed that the system data is reused until the convergence is achieved. The RL algorithm can learn the optimal control policy $u_{\tau i}^*$ from the real-time responses of the unknown environment to the original nonoptimal control policy u_{0i} . The performance function V_i^n and the control policy $u_{\tau i}^{n+1}$ can be approximated by two neural networks (NNs) with

$$\begin{aligned} \hat{V}_i^n(X_i) &= W_{1i} \vartheta_{1i}(X_i) \\ \hat{u}_{\tau i}^{n+1}(X_i) &= W_{2i} \vartheta_{2i}(X_i), i \in \Phi \end{aligned} \quad (25)$$

where $W_{1i} \in \mathbb{R}^{l_1}$ and $W_{2i} \in \mathbb{R}^{3 \times l_2}$ are the weights and $\vartheta_{1i}(X_i) \in \mathbb{R}^{l_1}$, $\vartheta_{2i}(X_i) \in \mathbb{R}^{l_2}$ the basis function of the NNs. Let $\Delta_{ui} = [\Delta_{ui}^1, \Delta_{ui}^2, \Delta_{ui}^3]^T = u_{0i} + u_{ei} - u_{\tau i}^n$ and

Algorithm 1: Data-Driven RL Algorithm.**1. Initialization**

Apply a fixed stabilizing policy u_{0i} and an exploring noise u_{ei} to the quadrotor system and collect the required information about the state and the control input. Start with arbitrary control policy $u_{\tau i}^0$.

2. Policy evaluation and policy improvement

For any given $u_{\tau i}^n$, one can obtain the performance function V_i^n and the control policy $u_{\tau i}^{n+1}$ simultaneously by solving the Bellman equation in (22).

If $\|u_{\tau i}^{n+1} - u_{\tau i}^n\| \leq \varepsilon_{\text{limit}}$, where $\varepsilon_{\text{limit}}$ is a small positive value, then stop and return $u_{\tau i}^{n+1}$; else let $n = n + 1$ and repeat 2.

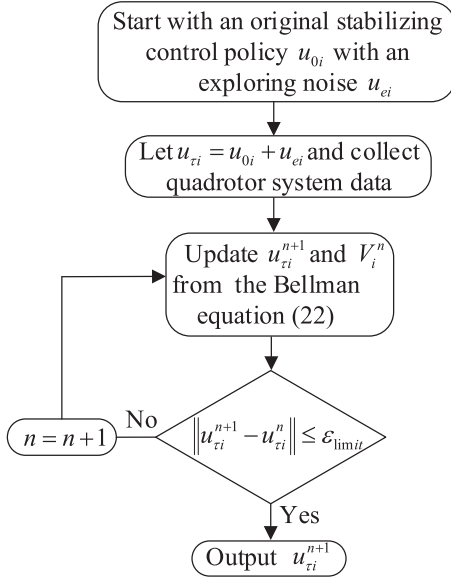


Fig. 1. Flowchart of Algorithm 1.

$R_{\Theta i} = \text{diag}(r_{i,1}, r_{i,2}, r_{i,3})$. Define $W_{2i,k}$ as the k th row of W_{2i} . Substituting (25) into (22), one can obtain that

$$\begin{aligned}
 e_{Bi}(t) &= W_{1i} \vartheta_{1i}(X_i(t + \Delta t)) - W_{1i} \vartheta_{1i}(X_i(t)) \\
 &\quad - \int_t^{t+\Delta t} -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} - (u_{\tau i}^n)^T R_{\Theta i} u_{\tau i}^n d\tau \\
 &\quad + 2 \sum_{k=1}^3 r_{i,k} \int_t^{t+\Delta t} W_{2i,k} \vartheta_{2i}(X_i(t)) \Delta_{u_i}^k d\tau \quad (26)
 \end{aligned}$$

where $e_{Bi}(t)$ is the approximation error and the weight matrices W_{1i} , W_{2i} can be updated by minimizing $e_{Bi}(t)$ through the least-square method under a persistence excitation condition (see [25]).

D. Stability Analysis

Let $\tilde{u}_{\Delta i} = \hat{u}_{\Delta i} - \bar{u}_{\Delta i}$ be the estimation error of the input uncertainty bound $\bar{u}_{\Delta i}$. Form (20), one can have that

$$\dot{\tilde{u}}_{\Delta i} = -k_r \sigma(t) (\tilde{u}_{\Delta i} + \bar{u}_{\Delta i}) + 2k_r \|R_{\Theta i} u_{\tau i}^*(x)\|. \quad (27)$$

Theorem 2: Consider the multiquadrotor system (14) controlled by the adaptive fault-tolerant controller (18). The tracking error $e_{\Theta i}$ of quadrotor i ($i \in \Phi$) will asymptotically converge to 0.

Proof: Consider the Lyapunov function as

$$L(V_i^*, \tilde{u}_{\Delta i}) = V_i^* + \frac{1}{2} k_r^{-1} \tilde{u}_{\Delta i}^2. \quad (28)$$

Differentiating (28) along (14) and using (17) yield

$$\begin{aligned}
 \dot{L}(V_i^*, \tilde{u}_{\Delta i}) &= (\Delta V_i^*)^T [F_{\Theta i}(X_i) + \bar{B}_i(u_{\tau i} + \rho(t - t_{si})u_{\Delta i}) + M_i \varepsilon_i] \\
 &\quad + k_r^{-1} \tilde{u}_{\Delta i} \dot{\tilde{u}}_{\Delta i} \\
 &\leq -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + \frac{1}{4} (\Delta V_i^*)^T \bar{B}_i R_{\Theta i}^{-1} \bar{B}_i^T \Delta V_i^* \\
 &\quad + \|(\Delta V_i^*)^T \bar{B}_i\| \bar{u}_{\Delta i} + (\Delta V_i^*)^T \bar{B}_i u_{\tau i} + k_r^{-1} \tilde{u}_{\Delta i} \dot{\tilde{u}}_{\Delta i}. \quad (29)
 \end{aligned}$$

Substituting (18) into (29), one can have

$$\begin{aligned}
 \dot{L}(V_i^*, \tilde{u}_{\Delta i}) &\leq (\Delta V_i^*)^T \bar{B}_i \left(\frac{u_{\tau i}^*(x)}{2} + \frac{\hat{u}_{\Delta i}^2 R_{\Theta i} u_{\tau i}^*(x)}{\|R_{\Theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta i} + \sigma(t)} \right) \\
 &\quad - e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + \frac{1}{4} (\Delta V_i^*)^T \bar{B}_i R_{\Theta i}^{-1} \bar{B}_i^T \Delta V_i^* \\
 &\quad + \|(\Delta V_i^*)^T \bar{B}_i\| \bar{u}_{\Delta i} + k_r^{-1} \tilde{u}_{\Delta i} \dot{\tilde{u}}_{\Delta i}. \quad (30)
 \end{aligned}$$

Because $u_{\tau i}^* = -\frac{1}{2} R_{\Theta i}^{-1} \bar{B}_i^T \Delta V_i^*$, one can obtain that

$$\begin{aligned}
 \dot{L}(V_i^*, \tilde{u}_{\Delta i}) &\leq -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + \frac{1}{4} (\Delta V_i^*)^T \bar{B}_i R_{\Theta i}^{-1} \bar{B}_i^T \Delta V_i^* \\
 &\quad + \|(\Delta V_i^*)^T \bar{B}_i\| \bar{u}_{\Delta i} - \frac{1}{4} (\Delta V_i^*)^T \bar{B}_i R_{\Theta i}^{-1} \bar{B}_i^T \Delta V_i^* \\
 &\quad - \frac{\hat{u}_{\Delta i}^2 (\Delta V_i^*)^T \bar{B}_i \bar{B}_i^T \Delta V_i^*}{2 \|R_{\Theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta i} + 2\sigma(t)} + k_r^{-1} \tilde{u}_{\Delta i} \dot{\tilde{u}}_{\Delta i} \\
 &\leq -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + 2 \|R_{\Theta i} u_{\tau i}^*(x)\| \bar{u}_{\Delta i} \\
 &\quad - \frac{2 \hat{u}_{\Delta i}^2 \|R_{\Theta i} u_{\tau i}^*(x)\|^2}{\|R_{\Theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta i} + \sigma(t)} + k_r^{-1} \tilde{u}_{\Delta i} \dot{\tilde{u}}_{\Delta i}. \quad (31)
 \end{aligned}$$

Substituting (27) into (31) yields

$$\begin{aligned}
 \dot{L}(V_i^*, \tilde{u}_{\Delta i}) &\leq -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + 2 \|R_{\Theta i} u_{\tau i}^*(x)\| \bar{u}_{\Delta i} + 2 \tilde{u}_{\Delta i} \|R_{\Theta i} u_{\tau i}^*(x)\| \\
 &\quad - \frac{2 \hat{u}_{\Delta i}^2 \|R_{\Theta i} u_{\tau i}^*(x)\|^2}{\|R_{\Theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta i} + \sigma(t)} - \sigma(t) \tilde{u}_{\Delta i} (\tilde{u}_{\Delta i} + \bar{u}_{\Delta i}). \quad (32)
 \end{aligned}$$

It can be noted that

$$\begin{aligned}
 2 \|R_{\Theta i} u_{\tau i}^*(x)\| \bar{u}_{\Delta i} - \frac{2 \hat{u}_{\Delta i}^2 \|R_{\Theta i} u_{\tau i}^*(x)\|^2}{\|R_{\Theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta i} + \sigma(t)} \\
 + 2 \tilde{u}_{\Delta i} \|R_{\Theta i} u_{\tau i}^*(x)\| = \frac{2 \sigma(t) \|R_{\Theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta i}}{\|R_{\Theta i} u_{\tau i}^*(x)\| \hat{u}_{\Delta i} + \sigma(t)} \leq 2 \sigma(t). \quad (33)
 \end{aligned}$$

From (32) and (33), one can have that

$$\begin{aligned}
 \dot{L}(V_i^*, \tilde{u}_{\Delta i}) &\leq -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + 2 \sigma(t) - \sigma(t) \tilde{u}_{\Delta i} (\tilde{u}_{\Delta i} + \bar{u}_{\Delta i}) \\
 &= -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + \frac{1}{4} \sigma(t) \left[-(2 \tilde{u}_{\Delta i} + \bar{u}_{\Delta i})^2 + 8 + \bar{u}_{\Delta i}^2 \right] \\
 &\leq -e_{\Theta i}^T Q_{\Theta i} e_{\Theta i} + \frac{1}{4} \sigma(t) (8 + \bar{u}_{\Delta i}^2). \quad (34)
 \end{aligned}$$

By integrating (34) from t_0 to t , one can obtain that

$$\begin{aligned} & \int_{t_0}^t e_{\Theta_i}^T Q_{\Theta_i} e_{\Theta_i} d\tau \\ & \leq L(V_i^*(t_0), \tilde{u}_{\Delta i}(t_0)) - L(V_i^*(t), \tilde{u}_{\Delta i}(t)) \\ & \quad + \frac{1}{4} (8 + \bar{u}_{\Delta i}^2) \int_{t_0}^t \sigma(t) d\tau. \end{aligned} \quad (35)$$

From (35), one can have that

$$\int_{t_0}^{\infty} e_{\Theta_i}^T Q_{\Theta_i} e_{\Theta_i} d\tau \leq L(V_i^*(t_0), \tilde{u}_{\Delta i}(t_0)) + \frac{1}{4} (8 + \bar{u}_{\Delta i}^2) \bar{\sigma} \quad (36)$$

and

$$\begin{aligned} & L(V_i^*(t), \tilde{u}_{\Delta i}(t)) \\ & \leq L(V_i^*(t_0), \tilde{u}_{\Delta i}(t_0)) + \frac{1}{4} (8 + \bar{u}_{\Delta i}^2) \int_{t_0}^t \sigma(t) d\tau. \end{aligned} \quad (37)$$

It can be seen from (37) that the errors e_{Θ_i} and $\tilde{u}_{\Delta i}$ are uniformly bounded. From (36) and the Barbalat Lemma (see [27]), one can obtain that $\lim_{t \rightarrow \infty} \|e_{\Theta_i}(t)\| = 0$. ■

IV. SIMULATION RESULTS

In this section, six quadrotor systems are modeled as (4) with the parameters as $J_{\Theta_i} = \text{diag}(4, 4, 8) 10^{-3} \text{kg} \cdot \text{m}^2$ and $b_{\tau_i} = \text{diag}\{41, 41, 110\}$ ($i \in \Phi$). A communication graph for the quadrotor team is given with $w_{16} = w_{21} = w_{32} = w_{43} = w_{54} = w_{65} = 1$ and $\gamma_1 = 1$. The quadrotor team is required to synchronize with a virtual leader to generate a sinusoidal trajectory. The initial states of the virtual leader and the following quadrotor team are given as $\Theta_0 = [50 \ -50 \ 15]^T$ deg, $\dot{\Theta}_0 = 0_{3 \times 1}$ deg/s, $\Theta_1 = [15 \ -15 \ 10]^T$ deg, $\dot{\Theta}_1 = [5 \ -5 \ 3]^T$ deg/s, $\Theta_2 = [10 \ -10 \ 6]^T$ deg, $\dot{\Theta}_2 = [3 \ -3 \ 2]^T$ deg/s, $\Theta_3 = [5 \ -5 \ 3]^T$ deg, $\dot{\Theta}_3 = [2 \ -2 \ 1]^T$ deg/s, $\Theta_4 = [-5 \ 5 \ -3]^T$ deg, $\dot{\Theta}_4 = [-2 \ 2 \ -1]^T$ deg/s, $\Theta_5 = [-10 \ 10 \ -6]^T$ deg, $\dot{\Theta}_5 = [-3 \ 3 \ -2]^T$ deg/s, $\Theta_6 = [-15 \ 15 \ -10]^T$ deg, and $\dot{\Theta}_6 = [-5 \ 5 \ -3]^T$ deg/s. The distributed observer is applied to the quadrotor with the parameter $\nu = 100$ and the initial states of the observers are arbitrarily selected as: $\varsigma_1 = [40 \ -55 \ 25 \ 5 \ -8 \ -5]^T$, $\varsigma_2 = [57 \ -48 \ 8 \ -7 \ -2 \ -10]^T$, $\varsigma_3 = [48 \ -54 \ 17 \ 4 \ -3 \ -6]^T$, $\varsigma_4 = [55 \ -48 \ 10 \ -2 \ 7 \ 6]^T$, $\varsigma_5 = [49 \ -53 \ 16 \ 3 \ -5 \ -6]^T$, and $\varsigma_6 = [47 \ -57 \ 22 \ 3 \ -8 \ -5]^T$. The Euler angle estimation errors $\tilde{x}_{\phi_i}$, $\tilde{x}_{\theta_i}$, and $\tilde{x}_{\psi_i}$ for quadrotor i are shown in Fig. 2. After the estimation errors converge to 0, the RL algorithm is applied for each quadrotor to iteratively learn the optimal control policy. An original proportional-derivative controller with an exploring noise is applied to the quadrotor team to generate system data. The exploring noise is designed as the sum of multiple singular signals. The time interval is chose as $\Delta t = 0.05$ s. The basis functions for the critic NN are chosen as multiple fourth-order polynomials and the basis functions for the actor NN are chosen as multiple first-order polynomials. It should be noted that the form of the NN is not restricted but sufficient number of NN is needed to approximate the optimal policies. The convergence of the weight matrices for the NNs is shown in Fig. 3. Then, the learned control policies are used to construct the fault-tolerant controller in (18) with $R_{\Theta_i} = I_{3 \times 3}$, $\sigma(t) = e^{-t}$, $\tilde{u}_{\Delta i}(0) = 0.1$, and $k_r = 0.1$. The actuator faults for the quadrotor team satisfy that $\chi_{11} = 0.12$, $\chi_{12} = 0.22$, $\chi_{13} = 0.31$, $\chi_{14} = 0.45$, $\chi_{21} = 0.43$, $\chi_{22} = 0.25$, $\chi_{23} = 0.34$, $\chi_{24} = 0.12$,

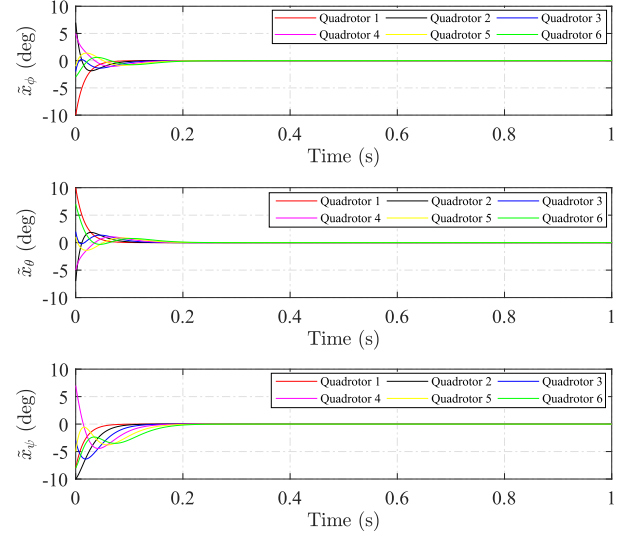


Fig. 2. State estimation errors of virtual leader for quadrotor team.

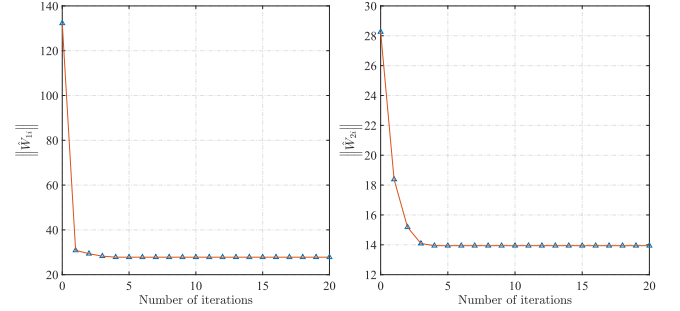


Fig. 3. Convergence of weights of performance function and control policy.

$\chi_{31} = 0.15$, $\chi_{32} = 0.32$, $\chi_{33} = 0.33$, $\chi_{34} = 0.41$, $\chi_{41} = 0.32$, $\chi_{42} = 0.22$, $\chi_{43} = 0.33$, $\chi_{44} = 0.14$, $\chi_{51} = 0.52$, $\chi_{52} = 0.34$, $\chi_{53} = 0.33$, $\chi_{54} = 0.44$, $\chi_{61} = 0.21$, $\chi_{62} = 0.22$, $\chi_{63} = 0.34$, and $\chi_{64} = 0.15$. Furthermore, the actuator faults of each quadrotor start at different times with $t_{s1} = 10$ s, $t_{s2} = 15$ s, $t_{s3} = 20$ s, $t_{s4} = 25$ s, $t_{s5} = 30$ s, and $t_{s6} = 35$ s. The attitude synchronization performances of the original controller and the learned fault-tolerant controller are shown in Figs. 4 and 5. One can see from Fig. 4 that the performance of the original controller is undesirable and even severe bias errors emerge when the actuator faults appear. One can observe from Fig. 5 that the flight performance is improved by using the learned fault-tolerant controller.

To provide comparison results, a robust compensator proposed in [9] is added to the original controller for the quadrotor team to restrain the multiple actuator faults. The quadrotor team is required to track a step signal to show the dynamic property of the quadrotor control system. The attitude tracking performances of one of the quadrotor team using the three controllers are shown in Fig. 6. The attitude tracking errors comparison of the three controllers is depicted in Fig. 7. As shown in Figs. 6 and 7, the multiple actuator faults occur at the time of 10 s and the attitude tracking performance of the original controller is improved by using the robust compensating approach in [9]. It can be seen from Figs. 6 and 7 that the proposed data-driven fault-tolerant controller exhibits a good attitude tracking performance among the three controllers.

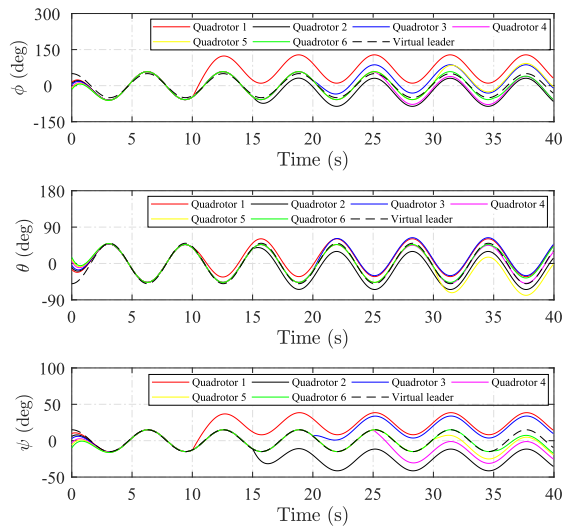


Fig. 4. Attitude synchronization of original controller for quadrotor team under actuator faults.

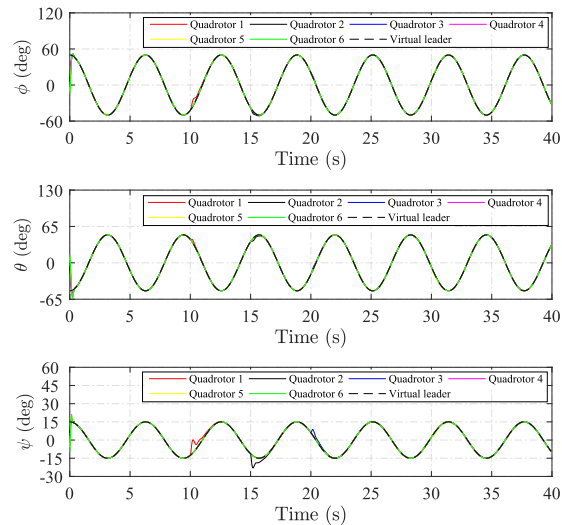


Fig. 5. Attitude synchronization of learned fault-tolerant controller for quadrotor team under actuator faults.

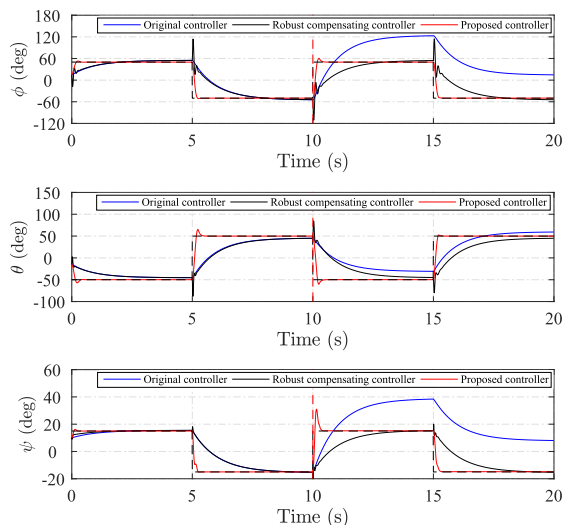


Fig. 6. Attitude tracking performance comparison of three controllers.

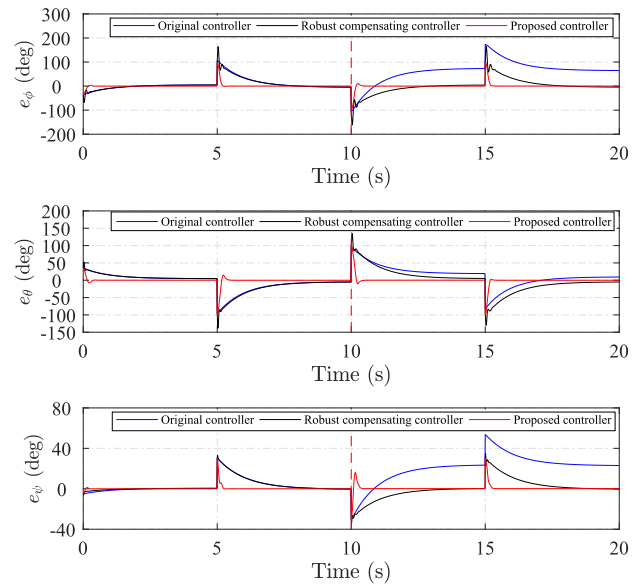


Fig. 7. Attitude tracking errors comparison of three controllers.

V. CONCLUSION

This article is intended to lay a preliminary foundation for the data-driven cooperative FTC of multiple quadrotors. The common partial LOE actuator faults in multiple quadrotors with nonlinear and coupled dynamics are considered. A distributed observer is constructed and an RL algorithm is designed to iteratively learn the optimal control policy without any knowledge of the quadrotor dynamics. Then, an adaptive fault-tolerant controller involving the optimal control policy is proposed to achieve the desired synchronization for the quadrotor team under multiple actuator faults. Stability of the proposed controller is proven and the simulation is conducted to show the effectiveness of the proposed controller.

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